Assignment-Regression

1. The problem comes under machine learning, supervised learning, Regression method

2. 6 rows, 1338 columns

3. Convert the String rows ("Sex, Smoker") to number (using the Nominal data)

To find following the machine learning regression method using in r2 value

1.MULTIPLE LINEAR REGRESSION (R^2 VALUE) = 0.789479

2.SUPPORT VECTOR MACHINE

S.NO	HYPER PARAMETER	LINEAR (R value)	RBF (R value)	POLY (R value)	SIGMOID (R value)
1	C=50	0.611637	0.144658	0.413598	0.409351
2	C=100	0.631243	0.313798	0.615655	0.531425
3	C=500	0.764481	0.661926	0.823464	0.444274

The SVM Regression **use R^2 value (nonlinear (Poly) and hyper parameter C (500) =** 0.823464

3. DECISION TREE:

S.NO	CRITERION	SPLITTER	R VALUE
1	squared error	best	0.689486
2	squared error	random	0.680082
3	friedman_mse	best	0.694359
4	friedman_mse	random	0.758841
5	Absolute_error	best	0.679620
6	Absolute_error	random	0.753571
7	poisson	best	0.705557
8	poisson	random	0.702897

The Decision Tree use R^2 value (Criterion = "friedman_mse", Splitter = random) = 0.758841

4.Random Forest

S.NO	CRITERION	MAX_FEATURES	N_ESTIMATORS	R_VALUE
1	*squared_error*	Sqrt	1000	0.872415
	squared_error	Sqrt	100	0.867975
	squared_error	Log2	1000	0.873246
2	*squared_error*	Log2	100	0.869963
3	Friedman_Mse	Sqrt	1000	0.871392
4	Friedman_Mse	Sqrt	100	0.869960
5	Friedman_Mse	Log2	1000	0.872601
6	Friedman_Mse	Log2	100	0.870928
7	absolute_error	Sqrt	1000	0.874263
8	absolute_error	Sqrt	100	0.872926
9	absolute_error	Log2	1000	0.874616
10	*absolute_error*	Log2	100	0.871916
11	*poisson*	Sqrt	1000	0.871508
12	*poisson*	Sqrt	100	0.871046
13	*poisson*	Log2	1000	0.872601
14	*poisson*	Log2	100	0.870935

The random Forest Regression R2 value (criterion='absolute error', max_features = log2, c=1000) = 0.874616

The final machine learning best method of Regression:

The random Forest Regression R2 value (criterion='absolute error', max_features = log2, c=1000) = 0.874616**.**